20261015 Quiz

for (MATH261) Discrete Mathematics

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Problem 1. Let $m, n \in \mathbb{N}$.

(a) Let $k_1, \dots, k_m \in \mathbb{N}$ with $k_1 + \cdots + k_m = n$. Prove that

$$\binom{n}{k_1, k_2, \dots, k_m} = \sum_{j=1}^{m} \binom{n-1}{k_1, \dots, k_j - 1, \dots, k_m}.$$

(b) Prove the multinomial theorem by induction on n: for arbitrary $x_1, \dots, x_m \in \mathbb{R}$,

$$(x_1 + \cdots + x_m)^n = \sum_{\substack{k_1+\cdots+k_m=n \\ k_1,\dots,k_m \in \mathbb{N}_0}} \binom{n}{k_1, \dots, k_m} x_1^{k_1} \cdots x_m^{k_m}.$$

Solution.

(a) Count the strings of length n containing exactly k_j copies of the symbol j for each $j \in [m]$. The total number of such strings is

$$\binom{n}{k_1, k_2, \dots, k_m}.$$

Classify these strings according to their last symbol. If the last symbol is j, then the first $n-1$ positions contain $k_j - 1$ copies of j and k_i copies of each $i \neq j$, giving

$$\binom{n-1}{k_1, \dots, k_j - 1, \dots, k_m}$$

strings. The m cases are disjoint and exhaustive, so summing them gives the desired identity.

(b) We use induction on n. For $n = 1$,

$$x_1 + \cdots + x_m = \sum_{j=1}^{m} x_j,$$

which is the multinomial formula in the base case.
Assume the formula holds for some $n \in \mathbb{N}$. Multiplying the induction hypothesis by $x_1 + \cdots + x_m$, consider a monomial

$$x_1^{\ell_1} \cdots x_m^{\ell_m}, \qquad \ell_1, \dots, \ell_m \in \mathbb{N}_0, \quad \ell_1 + \cdots + \ell_m = n + 1.$$

To obtain this monomial, we choose x_j from the additional factor, where $\ell_j \geq 1$; the term from the nth power must then have exponent $\ell_j - 1$ in x_j. Thus its coefficient is

$$\sum_{\substack{1 \leq j \leq m \\ \ell_j \geq 1}} \binom{n}{\ell_1, \ldots, \ell_j - 1, \ldots, \ell_m}.$$

Using the factorial formula for multinomial coefficients and $\ell_j! = \ell_j(\ell_j - 1)!$ for $\ell_j \geq 1$, we obtain

$$\begin{aligned}
\sum_{\substack{1 \leq j \leq m \\ \ell_j \geq 1}} \binom{n}{\ell_1, \ldots, \ell_j - 1, \ldots, \ell_m} &= \sum_{\substack{1 \leq j \leq m \\ \ell_j \geq 1}} \frac{n!}{\ell_1! \cdots (\ell_j - 1)! \cdots \ell_m!} \\
&= \frac{n!}{\ell_1! \cdots \ell_m!} \sum_{j=1}^{m} \ell_j \\
&= \frac{(n+1)!}{\ell_1! \cdots \ell_m!} \\
&= \binom{n+1}{\ell_1, \ldots, \ell_m}.
\end{aligned}$$

Here the indices with $\ell_j = 0$ contribute zero to the sum after factoring. Therefore the multinomial formula holds for $n+1$, and hence for every $n \in \mathbb{N}$ by mathematical induction.

□

Problem 2. There are 10 couples, each with one child. The 10 children are lost in a dark cave, and each couple retrieves exactly one child, so that every child is retrieved once. How many assignments pair at least one child with their own parents?

Solution 1. Label the couples and their children by [10], with child i belonging to couple i. An assignment of the children to the couples is a permutation of [10]. For $i \in [10]$, let A_i be the set of assignments in which couple i retrieves child i.

If k specified couples are correctly paired, then the remaining $10 - k$ children may be assigned arbitrarily to the remaining couples, giving $(10 - k)!$ assignments. Hence, by inclusion-exclusion,

$$\begin{aligned}
|A_1 \cup \cdots \cup A_{10}| &= \sum_{k=1}^{10} (-1)^{k+1} \binom{10}{k} (10 - k)! \\
&= 10! \sum_{k=1}^{10} \frac{(-1)^{k+1}}{k!} \\
&= 2293839.
\end{aligned}$$

□

Solution 2. Every assignment is a permutation of [10]. Let D_{10} be the number of permutations with no fixed points. The desired assignments are precisely those that are not derangements. Using the derangement formula proved in Problem 5(a), their number is

$$\begin{aligned}
10! - D_{10} &= 10! - 10! \sum_{k=0}^{10} \frac{(-1)^k}{k!} \\
&= 2293839.
\end{aligned}$$

□

Problem 3. Find the number of integer solutions of

$$x_1 + x_2 + x_3 + x_4 = 15$$

satisfying

$$2 \le x_1 \le 6, \qquad -2 \le x_2 \le 1, \qquad 0 \le x_3 \le 6, \qquad 3 \le x_4 \le 8.$$

Solution. Set

$$y_1 = x_1 - 2, \qquad y_2 = x_2 + 2, \qquad y_3 = x_3, \qquad y_4 = x_4 - 3.$$

Then

$$y_1 + y_2 + y_3 + y_4 = 12,$$

with

$$0 \le y_1 \le 4, \qquad 0 \le y_2 \le 3, \qquad 0 \le y_3 \le 6, \qquad 0 \le y_4 \le 5.$$

Without the upper bounds, stars and bars gives

$$\binom{12+4-1}{4-1} = \binom{15}{3}$$

nonnegative solutions.

Among these nonnegative solutions, let A_i be the set of solutions violating the upper bound on y_i. These sets have sizes

$$|A_1| = \binom{10}{3}, \quad |A_2| = \binom{11}{3}, \quad |A_3| = \binom{8}{3}, \quad |A_4| = \binom{9}{3}.$$

The nonzero pairwise intersections have sizes

$$|A_1 \cap A_2| = \binom{6}{3}, \quad |A_1 \cap A_3| = \binom{3}{3}, \quad |A_1 \cap A_4| = \binom{4}{3},$$
$$|A_2 \cap A_3| = \binom{4}{3}, \quad |A_2 \cap A_4| = \binom{5}{3}.$$

All other pairwise intersections and all intersections of three or more of the A_i are empty. Therefore, by inclusion-exclusion, the required number is

$$\begin{aligned}\binom{15}{3} &- \left(\binom{10}{3} + \binom{11}{3} + \binom{8}{3} + \binom{9}{3}\right) \\ &+ \left(\binom{6}{3} + \binom{3}{3} + \binom{4}{3} + \binom{4}{3} + \binom{5}{3}\right) \\ &= 69.\end{aligned}$$

□

Problem 4. Let $n \in \mathbb{N}$ and $k \in \mathbb{Z}$ with $0 \le k \le n$. Prove that the number of k-element subsets of $[n]$ containing no two consecutive integers is

$$\binom{n-k+1}{k}.$$

We use the convention

$$\binom{a}{b} = 0 \qquad (a, b \in \mathbb{N}_0,\ b > a).$$

Solution. The case $k = 0$ is immediate. Assume $k \geq 1$, and let $A = \{a_1, \ldots, a_k\}$ be such a subset, with

$$1 \leq a_1 < a_2 < \cdots < a_k \leq n.$$

Since no two selected integers are consecutive,

$$a_{i+1} \geq a_i + 2$$

for every $1 \leq i < k$. Define

$$b_i = a_i - (i-1), \qquad 1 \leq i \leq k.$$

Then

$$1 \leq b_1 < b_2 < \cdots < b_k \leq n - k + 1,$$

so $\{b_1, \ldots, b_k\}$ is a k-element subset of $[n-k+1]$.

Conversely, from any

$$1 \leq b_1 < b_2 < \cdots < b_k \leq n - k + 1,$$

defining $a_i = b_i + (i-1)$ gives a k-element subset of $[n]$ with no two consecutive elements. These constructions are inverse to each other. Hence the desired subsets are in bijection with the k-element subsets of $[n-k+1]$, and their number is

$$\binom{n-k+1}{k}.$$

□

Remark. The cyclic analogue, known as *Cayley's problem*, asks for the number of ways to select k vertices of a regular n-gon with no two selected vertices adjacent (including the first and last vertices). For $n \geq 3$ and $0 \leq k \leq \lfloor n/2 \rfloor$, the number of selections is

$$\frac{n}{n-k}\binom{n-k}{k}.$$

For $k > \lfloor n/2 \rfloor$, there are no such selections.

Problem 5. Let $n \in \mathbb{N}$. An n-derangement is a permutation $\pi \in S_n$ with no fixed point, i.e.

$$\pi(i) \neq i \qquad \text{for every } i \in [n].$$

Let D_n denote the number of n-derangements.

(a) Prove that

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

(b) Prove that, for every $n \geq 2$,

$$D_n = nD_{n-1} + (-1)^n.$$

Solution.

(a) For each $i \in [n]$, let

$$A_i = \{\pi \in S_n \mid \pi(i) = i\}.$$

The permutations that are not derangements are precisely the elements of $A_1 \cup \cdots \cup A_n$. For any distinct indices $i_1, \ldots, i_k$,

$$|A_{i_1} \cap \cdots \cap A_{i_k}| = (n-k)!,$$

since those k points are fixed and the remaining $n-k$ elements may be permuted arbitrarily. Hence inclusion-exclusion gives

$$\begin{aligned}|A_1 \cup \cdots \cup A_n| &= \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} (n-k)! \\ &= n! \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!}.\end{aligned}$$

Therefore

$$\begin{aligned}D_n &= n! - |A_1 \cup \cdots \cup A_n| \\ &= n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.\end{aligned}$$

(b) Using part (a),

$$\begin{aligned}nD_{n-1} &= n! \sum_{k=0}^{n-1} \frac{(-1)^k}{k!} \\ &= n! \left(\sum_{k=0}^{n} \frac{(-1)^k}{k!} - \frac{(-1)^n}{n!} \right) \\ &= D_n - (-1)^n.\end{aligned}$$

Rearranging yields

$$D_n = nD_{n-1} + (-1)^n.$$

□

Remark. Another standard recurrence for derangement numbers is

$$D_n = (n-1)(D_{n-1} + D_{n-2}), \qquad n \geq 3.$$

For every $n \geq 1$, D_n is the nearest integer to $n!/e$:

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

The classical *hatcheck lady problem* asks for the probability that no one receives their own hat when n hats are randomly assigned. Under a uniformly random assignment, this probability is $D_n/n!$, and

$$\lim_{n \to \infty} \frac{D_n}{n!} = \frac{1}{e}.$$